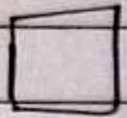


Date _____

LECTURE - 13

Wednesday
7/10/26!



Max. Sub Array (using Dynamic Programming) ..

Sub array is a continuous part of array with min length = 1
& max length = len(array).

$O(n^2)$ = Fix one element and compare with others to find max sum.
Repeat it for all elements by fixing each i & $j = i + \dots$

Bottom Up $\rightarrow O(n)$: named 'meow'

- (creates) an array of same length as original (A)
- Each index i of meow will either have:
 - 1 - Sum of solution found till $(i-1)^{th}$ index
 - 2 - start a new sum from index ' i ' -

OR } max.
of these two

$$\text{Meow}[i] = \max(\text{Arr}[i], \text{meow}[i-1] + \text{Arr}[i])$$

Date _____

- Another thing to keep a check of is the index the sub-array with max sum started.
- for that we'll keep a track with another array 'S'.
- Each time we'll be using the sum till previous index; the value of that index in S will remain same (i.e. where the sub array started from).
- Whenever we start a new sum from i^{th} index; the value in 'S' array's i^{th} index will change to i & it will continue in S's $i+1, i+2 \dots$ index till a new sum is initiated.

Q.

1	2	3	4	5	6	7	8	9	10	11	12
15	-11	-7	2	3	7	-5	13	-2	-5	8	-5

 Arr

$i=2$
 $\max(\text{Arr}[2], \text{Meow}[1] + \text{Arr}[2])$

$\max(-11, 15 + (-11))$

$\max(-11, 4) = 4$

$i=8$
 $\max(13, 13 + 7)$

$= 20$

Meow

1	2	3	4	5	6	7	8	9	10	11	12
15	4	-3	2	5	12	7	20	18	13	21	16

S

1	1	1	4	4	4	4	4	4	4	4	4
---	---	---	---	---	---	---	---	---	---	---	---

means that the sum of sub array start from index = 1 in sub array Meow[1], Meow[2], Meow[3].

The max. sub array till the end is winner & it's long chain has started from index = 4 of Meow array.

NOTE : Since we just need the previous sum if going with option 1; therefore this could be done with variables in space complex $O(1)$
 → Home work 3 solution